Masih Tanoursaz

Cloud Computing & Intelligent Systems Research

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Research Profile

I'm a Computer Engineering graduate interested in sustainable cloud computing, intelligent resource allocation, and applied AI. My bachelor's research compared reinforcement learning policies with LLM-based control agents in SustainDC, examining energy, carbon, water use, and system behavior. I bring practical cloud and model-serving experience to research, alongside teaching in networking, IT project management, technical communication, and embedded systems.

Research Interests

Cloud and edge computing; resource allocation and scheduling; sustainable data centers; reinforcement learning and LLM agents for infrastructure control; applied AI and data-driven decision support.

Education

Bachelor of Science in Computer Engineering

2021 - 2025

Isfahan University of Technology

- GPA: 89/100.
- Bachelor's final project: evaluating LLM-based and reinforcement learning agents for sustainable data center optimization.
- Relevant study: networks, operating systems, databases, artificial intelligence, software engineering, and IT project management.

Research Experience

Intelligent Agents for Sustainable Data Center Optimization

2025

Bachelor's research project | Isfahan University of Technology

- Designed a comparative experimental workflow covering workload shifting, data-center cooling, and battery charge/discharge decisions.
- Compared pretrained multi-agent reinforcement learning policies with LLM-based control agents.
- Built a pipeline that translated numerical observations into structured prompts and validated JSON actions before applying them to the simulator.
- Evaluated energy consumption, carbon footprint, water usage, cumulative reward, and agent-level returns.
- Worked with multi-agent RL methods including HAA2C, HAPPO, HASAC, and MAPPO, with reproducible experiment and analysis workflows.

Research output: undergraduate project and report.

Research & Technical Skills

AI Research & Evaluation: Reinforcement learning | Multi-agent systems | LLM agents | SustainDC | Experimental planning | Data analysis

Programming & Software Development: Python | Go | Bash | C/C++ | SQL | HTML | CSS | Qt | Git

Data Systems & Analytics: PostgreSQL | Microsoft SQL Server | Data warehousing | ETL | Dimensional modeling | Power BI | Excel

Cloud & Delivery: AWS | Kubernetes | Docker | Terraform | GitHub Actions | Jenkins | Linux | Git

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Cloud Computing & Intelligent Systems Research | Continued

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Teaching Experience

Teaching Assistant | Computer Engineering Mar 2024 - Jan 2026
Isfahan University of Technology | Isfahan, Iran

- Supported IT Project Management, Fundamentals of IT, Computer Networks, Network Laboratory, Scientific and Technical Presentation, Microprocessor, and Digital Systems Design Laboratory.
- Head Teaching Assistant for Scientific and Technical Presentation: coordinated course workflows and reviewed technical documents and presentations.
- Guided Cisco networking labs covering VLANs, routing, switching, connectivity testing, and troubleshooting.
- Supported project planning, documentation, scheduling and risk analysis; assisted AVR ATmega32 and Proteus exercises, grading, mentoring, and student project evaluation.

Selected Technical Projects

Flight Reservation Data Warehouse Jun 2025 - Aug 2025
Database II | Data engineering

- Designed transactional, staging, and dimensional warehouse layers for flight reservation data.
- Wrote SQL and Python scripts to generate test data, transform records, and load the warehouse.

Local AI Inference & Computer Vision Sep 2026 - Present
Personal AI / MLOps projects

- Served Qwen3:8B through vLLM and connected the model endpoint to personal projects.
- Used NVIDIA Triton to serve YOLOv12 for object-detection experiments, inspecting visual inputs and model outputs.

Industry & Product Experience

Cloud / Backend Development Intern Jun 2024 - Oct 2024
Payam Pardaz | Sadra Cloud Team | Isfahan, Iran

- Developed and tested containerized Go services; worked with HashiCorp Vault, MOSN, Linux, and service-to-service communication in the Sadra Cloud R&D team.

Product Innovator | MED-X 2025
MED-X Health & Innovation Program

- Co-developed an early-stage digital health concept addressing antibiotic resistance; contributed user and market analysis, value-proposition and business-model work, and a final pitch to judges and mentors.

Academic Service & Leadership

Public Relations Coordinator (Feb 2024 - May 2025): Public relations, outreach, and sponsorship coordination for university technical events.

Registration Manager | Sisoog (Nov 2024): Managed registration and check-in for 870+ participants.

Co-Head | APACHEE Club (Mar 2024 - Oct 2024): Coordinated cybersecurity events and CTF activities; secured 35+ million Tomans in sponsorship.

Branding Manager (Feb 2023 - Jan 2024): Led branding and promotion and built a Telegram bot for event messages.

Executive & Branding Team Member (Jan 2023 - Feb 2023): Supported promotion of a programming competition with 300+ participants.

Languages: Persian and English.